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A STUDY ON SUPPLY CHAIN COODINATION AND ITS IMPACT ON BUSINESS PERFORMANCE

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ABSTRACT:

In order to make the most of supply chain, there must be the element of coordination within supply chain partners; for efficiency in both manufacturing and service business the strong supply chain partners co-ordination plays a crucial role. "Supply chain coordination and its effects on business performance: An exploratory investigation into the effect of information sharing, collaboration and communication."

"This research uses a combination of primary (questionnaire-based survey of 50 people) and secondary sources for data gathering. The study measures business performance using the dependent variables of cost efficiency, service quality, operational efficiency and customer satisfaction." In this study, the independent variables include coordinating processes that may affect the dependent variables. The study will apply descriptive statistics on the gathered data.

"The findings of this study are that there is an expected improvement in business performance through high coordination because of improved service quality, operational efficiency and customer satisfaction. On the other hand, poor coordination affects business performance through delayed deliveries and stockouts."

Keywords: *Supply Chain Coordination, Business performance, Operational efficiency, Information Sharing level, Customer Satisfaction.*

INTRODUCTION

“In today’s world economy, organizations become increasingly dependent on one another through networking or what is popularly termed as supply chain management. This calls for coordinated planning regarding procurement, production and delivery of goods/services among different supply chains.” Due to stiff competition and consumer demands, there are demands for increased efficiency and performance from the organizations. This is the epidemiology in which the importance of supply chain coordination has emerged as an important organizational skill required for its survival in the long-run.

Supply chain coordination refers to the extent to which different functions, suppliers, manufactures and customers coordinate their processes, decisions and information flows. “Coordination of these processes eliminates waste of resources leading to improved efficiency, increased services and reduced costs. The lack of coordination leads to delays, unnecessary inventories and dissatisfied customers.” While in emerging economies such as India, the issue of legislative coordination in different sectors arises due to rapid industrialization and business models have started adopting technology, yet at the same time there are issues of fragmentation of supply networks and inadequate IT connectivity.

However, it is crucial since, in practice, the way of achieving coordination is rather ineffective and the gap between theory and practice is apparent. “Consequently, this study aims at investigating the effect of supply chain coordination on the performance of businesses through such aspects as information exchange, cooperation, and IT utilization, addressing key research issues regarding what coordination means and what difficulties companies face.”

This research follows these research questions: (1) What is the influence of supply chain coordination on business performance? (2) Which coordination mechanisms do the best job of enhancing organizational performance? and (3) What difficulties do firms face in attaining effective coordination? Answering these questions will present insights on the practical effectiveness of coordination strategies between business projects in professional environments.

“This paper shall then be organized in the following structure. In the first section after this introduction, literature review and research gap will be explained. In the subsequent section, methodology of this research will be discussed in relation to the methods employed for data collection and data analysis.” In the following section, the presentation and analysis of data gathered as well as findings of this study will be explained, followed by the practical implications of this finding. The paper will then conclude with summary of findings and areas for future research.

Literature Review

“Efficient coordination of the supply chain has always been an important aspect for a successful supply chain operation.” In earlier research works, it was considered that one of the important aspects which contribute to excellence in the supply chain would be the coordination of different functions in the supply chain. Efficient coordination can be defined as "the processes of the supply chain working together and that there is no wasted effort due to responsibility fragmentation." Similarly, Mentzer et al. Williams et al. (2001) define supply chain management as the systemic coordination of such business functions as marketing, sales, product distribution, and logistics across different firms that participate in the

enrichment process; they emphasize integration among firms as an enabler to increase responsiveness in meeting customer needs and thus attain firm strategic goals.

One of the most talked about notion in the field of supply chain coordination is the bullwhip effect given by Lee, Padmanabhan, and Whang (1997). They show that inadequate coordination and insufficient information sharing can create demand distortion throughout the supply chain, leading to wasteful behavior such as overstocking and proliferation of excess cost. Later research has highlighted the importance of proper information sharing as an effective way to soften the bullwhip effect and enhance forecasting accuracy (Simatupang & Sridharan, 2005).

Another factor cited as an important performance driver is collaboration between supply chain partners. Cao and Zhang (2011) posit that joint decision-making and sharing of information or resources as collaborative practices increase operational efficiency while providing mutual benefits for all parties involved. Likewise, Flynn et al. (2010) report that greater supply chain integration is positively linked to firm performance, especially with respect to operational and financial outcomes. What do these studies tell me about achieving competitive advantage?

The coordination in a supply chain can perennially be improved through technological advancements. The use of information technology like ERP systems helps the companies through enhanced visibility and coordination of the supply chain network (Christopher, 2016). However, these technologies, even if they are in place, still a lot of companies especially the ones located in developing countries face problems adopting the same due to fund constraint, lack of necessary technical know-hows etc.

While most of the studies on the link between supply chain coordination and firm performance conclude on positive correlation, few researches however present contrary findings for certain issues. For example, Fabbe-Costes and Jahre (2008) observe that the gains from integration would not be uniformly realized across industries or degree of implementation. Similar arguments have been extended by Barratt (2004) by adding the role of trust and alignment of cultures for successful coordination.

In the case of India, research conducted on supply chain coordination are few and disjointed. Although some studies highlight increasing application of supply chain coordination practices by Indian firms, however there is paucity of empirically testable literature concerning the performance gains achieved by them. Also, while many research endeavors explore supply chain coordination in the field of disaster relief, but much of the literature confines the issue to only specific aspects of coordination (i.e. Logistics and technology) and provides limited holistic viewpoint of the multiple dimensions of coordination.

Research Gap

Another area lacking is perception-based studies to show how people from different segments ranging from students to top level managers to entrepreneurs look at issues. Past research conducted has been largely using firm-level or industry specific data (which may not be able to adequately handle the real world supply chain coordination context) and hence an extended awareness with respect to such context has been built. As understanding of supply chain coordination and application at developing economies is yet to be developed further, more study needs to be conducted to address how perceptions differ across groups for the use of practices and the resulting impact on performance. The above review makes it clear that while much progress has been achieved with respect to our understanding of supply chain

coordination, a vacuum is still present in the literature for contexts such as emerging markets like India. This paper attempts to fill this gap by offering an integrative perspective on all possible coordination mechanisms and their implications for business performance, thus contributing both to academic literature and managerial practice.

RESEARCH DESIGN AND METHODOLOGY

This research is quantitative-based to study the relationship between supply chain coordination and business performance. This is a descriptive study combining analytical tools; the purpose of this research is to describe how coordination occurs in organizational contexts and analyze its influence on performance results. The prevailing sentiment suggests a quantitative approach for this study, as it enables the operationalization of variables and the utilization of statistical techniques to objectively test proposed hypotheses (McBurney and White, 2004).

Cross sectional design used wherein data from the respondents were gathered at one particular instance in time. It enables us to collect up-to-date information on coordination methods currently employed by firms and sheds some light on their impact on business performance. Respondents from varied segment groups are included in the study so that we can obtain a 360-degree insight on coordination practices so we can illustrate how such practices influence other firms.

Data Collection

Data is either primary or secondary. Primary data were gathered using a thorough questionnaire designed on the Google Forms application. It used close-ended question responses on a 5-point scale where: Strongly disagree, disagree, neutral, agree, strongly agree was the order. “This question was intended to get answers for the particular factors involved in supply chain coordination including sharing of information among partners, cooperation among partners, technology use among partners, and extent of trust among partners.”

“The responses gathered through the google form questionnaire were 50 in number. They included people from diverse categories who were working professionals, unemployed, students, and entrepreneurs, most of whom belonged to Raipur Chhattisgarh.” Such diversity in respondent profiles ensures that a wider range of perspectives related to coordination and its impact on business performance is obtained. Respondents who may not belong to the supply chain industry have knowledge of various other businesses which aided their evaluation.

Secondary data in the form of previous studies conducted in similar topics along with textbooks, journals, and papers published by companies has been used to formulate the theoretical basis for the study.

Sample Design

The study employed a convenience sampling technique for its sample because of time and access constraints. Respondents were selected based on availability and willingness to complete the survey. Despite that, this method gives an overall indicative picture of the things being studied and although might not reach full representability is widely used in academic research and is very suitable for exploratory studies like this one.

We have balanced the sample with its distribution according to professionals, students and business people. The objective here is to find the perceptions and knowledge about role and advantages of Supply Chain Coordination in business performance.

Variables of the Study

The linkage of these variables was examined within the framework of the research by assessing the positive influence of supply chain coordination on business performance. “Each of these variables (information sharing, distribution, technology integration, and trust) would have been aligned with supply chain coordination strategy if these factors were to be classified as the independent variables that form the basic structure for successful supply chain coordination.”

“Business Performance measures such as costs, efficiency, service delivery, operations, and customer satisfaction form business performance.” These are largely responsible for determining the level of success within an organization. They were proportional to the quality of supply chain coordination.

Hypotheses

The following hypotheses are formulated based on an extensive study of existing literature for the study:

H1: Coordination in the supply chain has a positive impact on the performance of businesses.

H2: Satisfaction of customers is positively influenced by the cooperation of supply chain partners.

Data Analysis Techniques

The questionnaire responses went through very simple descriptive statistical analytical techniques. “Methodologically, analysis of data was done through calculation of percentage in summary of the collected data and identification of common trends in respondents. Frequency tables Validation: The collected data was transformed into frequency tables for the better presentation of the analysis.” We were examining the view of respondents on their business supply chain coordination and relating it to business performance. From the trends identified, I derived some of my key takeaways.

The analyses result are presented using tables for clearer explanations. Such methods permit organizations to make reasoned generalizations concerning how supply chain collaboration impact the organization's performance.

DATA ANALYSIS & FINDINGS

This part of the thesis is focusing on the primary data that have been collected through questionnaire survey, using Google forms. The survey had collected from respondents from all walks of life and business, from all types of professions, students, employed, unemployed including businessmen and it comprised of 50 people. “Analyzed the business data by the calculation of percentage and frequency distribution in the aim to obtain how business performance affects by the supply chain coordination.”

Demographic Analysis

Category	Sub-category	Number of Respondents
Age	21–30 years	33
	31–40 years	18
Gender	Female	30
	Male	21
Occupation	Employed	23
	Students	17
	Unemployed	7
	Business Owners	4

Awareness of Supply Chain

Response	Number of Respondents
Very familiar	24
Somewhat familiar	20
Heard of it	7

Importance of Coordination

Response	Frequency
Agree	37
Strongly Agree	10
Neutral	4

Delivery Performance

Response	Frequency
Often	38
Always	9
Sometimes	4

Stock Availability

Response	Frequency
Often	30
Sometimes	15
Always	5
Rarely	1

Impact of Coordination on Performance

Statement	Agree/Positive Responses
Better coordination reduces stock shortages	39
Coordination improves service quality	35
Coordination improves customer satisfaction	44
Coordination improves business performance	42

Role of Technology

Response	Frequency
Agree	37
Strongly Agree	3
Neutral	10
Strongly Disagree	1

Customer Satisfaction and Loyalty

Statement	Positive Responses
Happier when delivery is on time	44
More trust in reliable companies	40
More repurchase behavior	41

Overall Rating

Rating (Out of 5)	Frequency
4	39
5	7
3	5

Key Findings

The study of the data we collected provides invaluable insights about the influence of supply chain coordination on business performance. Seems to be many of the interviewees were aware of concepts of the supply chain, this gives us some idea that coordination seems to have an appropriate awareness. From this, it is seen that it is becoming more and more popular among different groups of people and scholars as well as students are all working toward the field of supply chain management. It was shown that many of the data concentrated on the part that partners in the supply chain unnecessarily coordinating with each other. Most of the respondents also agree that good coordination between supplier, manufacturer, delivery system, etc. Is necessary for the business, and poor coordination is the critical cause of delays in order and poor performance. One main key finding focused on inventory & stock management; many of the respondents state that it sometimes does not have enough stock, which shows that there is likely some coordination gap especially regarding to demand forecasting and inventory management. This information added that effective information links between business partners are needed.

Results also show a high correlation between supply chain coordination and KPIs. Respondent consensus: Better coordination leads to better quality of service, operational efficiency and customer satisfaction. Promotion of coordination practices are vital in explaining the better performance of organizations, as highlighted in this research.

There was also clear acknowledgment in the theme around technology and its role to increase coordination in the supply chain. The majority of the respondents all concurred better coordination with faster communications and real-time information sharing happened through digital tools/ systems. “However, despite all this, a neutral answer may also have its own merits that can help in the broader application and optimal use of technology.”

As revealed by consumer behavior research, it also places emphasis on the timely availability of goods and reliability, which is a very important aspect in determining consumer satisfaction, trust, and purchasing intention. “As such, it stands to reason that the more coordination exists among businesses in delivering their products/services, the more likely will be the satisfaction and loyalty of customers towards these firms.” Hence, coordination serves as the basic building block of lasting customer relationships.

“On the whole, this study provides sufficient evidence for acknowledging the fact that the coordination of the supply chain has a positive effect on business performance.” Focusing on coordination helps the organization achieve customer satisfaction, effectiveness and competitiveness.

Key Implications

There are interesting implications that arise from these observations in terms of management. “This means that the organizations need to focus on the improvement of the coordination processes among the participants in the supply chains, through the enhancement of communication, cooperation and exchange of information.” These factors will ensure reduction in the uncertainties in the decision-making processes.

One of the interesting management issues that can be derived from these observations is the importance of having good relationships between the partners and establishing trust. “There are "how-to" practices, but it is vital to ensure that long-term, reciprocal and transparent relationships are established with the partners. These practices will ensure higher efficiency in supply chain coordination.”

There are new technological approaches that need to be considered to facilitate better coordination within the supply chain. The adoption of new information technology approaches will be useful in ensuring that there is better access to information and increase in supply chain visibility.

CONCLUSION

“From the findings obtained during the course of this study, it can be clearly seen that there is clear evidence indicating that coordination of the supply chain process has great effects on the overall performance of a business entity. The majority of respondents agree that coordination between suppliers and the firm as well as coordination among delivery processes is key to the success of their organizations’ operations.” The high level of comprehension of the concept of supply chain coordination from the responders suggests that the importance of the former on the business world is growing.

Poor coordination among all actors in the supply chain are major cause of rain delays and inefficiencies as find from the result. An enormous number of respondents believe that lack of coordination among parties result in various problems such as late delivery of orders and shortage of supply. This observation is consistent with what had been mentioned by various scholars on variation in demand and inefficiencies associated with poor information sharing within the supply chain; stock shortages are consistent with the need for better integration and matching of the demand and supply throughout the supply chain network.

An encouraging result that emerged from this research is the positive association of coordination on the performance of business operations. In fact, a huge majority of responders feel that coordinating with multiple providers leads to greater customer satisfaction (92%), improved operational efficiency (90%) and quality of service (91%). The findings align well with existing literature in that more collaboration and integration of partners involved in the supply chain operations tends to yield superior performance from the organization. It is of interest that there is so much accord on customer satisfaction, suggesting a fairly immediate impact of coordination on the business-customer relationship.

Respondents recognize the significant effect of technology in the coordination of the supply chain activities. All participants agree with that new technology enhance coordination by improve communication channels and rapid flow of information. Such perception is consistent with literature emphasizing the role of technology in supply chain. There are some who remain neutral in regards with

adopting technology in coordinating supply chain; this imply that the adoption of technology throughout supply chain has yet to reach uniformity in various organization.

It is clear from result that how coordination is relating to customer choice. The respondents stated that coordination builds trust which encourage the repeat purchase behavior among the customers and long term relationship. Hence it is clearly indicating that coordination is not just a factor to improve internal efficiency, but also important for better customer relationships as expected performance can be delivered through well-coordinated supply chain operations; and for the long term sustained performance of the business.

The findings, obtained from this research, support, and are therefore consistent with previous research/literature and the hypotheses proposed. Supply chain coordination has been shown to have a direct positive effect on businesses by enhancing efficiency (reducing operating problems) and improving customer satisfaction as well as positively impacting supply chain coordination to the overall success of the business enterprise.

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